

- ```
Clone de Mininet-VM [Executando] - Oracle VirtualBox
```
- Arquivo Máquina Visualizar Entrada Dispositivos Ajuda
- ```
mininet@mininet-vm:~$ sudo mn --topo linear,8 --mac --link tc,bw=20  
*** Creating network  
*** Adding controller  
*** Adding hosts:  
h1 h2 h3 h4 h5 h6 h7 h8  
*** Adding switches:  
s1 s2 s3 s4 s5 s6 s7 s8  
*** Adding links:  
(20.00Mbit) (20.00Mbit) (h1, s1) (20.00Mbit) (20.00Mbit) (h2, s2) (20.00Mbit) (20.  
(20.00Mbit) (h3, s3) (20.00Mbit) (20.00Mbit) (h4, s4) (20.00Mbit) (20.00Mbit) (h5,  
s5) (20.00Mbit) (20.00Mbit) (h6, s6) (20.00Mbit) (20.00Mbit) (h7, s7) (20.00Mbi  
t) (20.00Mbit) (h8, s8) (20.00Mbit) (20.00Mbit) (s2, s1) (20.00Mbit) (20.00Mbit)  
(s3, s2) (20.00Mbit) (20.00Mbit) (s4, s3) (20.00Mbit) (20.00Mbit) (s5, s4) (20.  
(20.00Mbit) (20.00Mbit) (s6, s5) (20.00Mbit) (20.00Mbit) (s7, s6) (20.00Mbit) (20.0  
Mbit) (s8, s7)  
*** Configuring hosts  
h1 h2 h3 h4 h5 h6 h7 h8  
*** Starting controller  
c0  
*** Starting 8 switches  
s1 s2 s3 s4 s5 s6 s7 s8 ... (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.0  
Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (  
(20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mi  
t) (20.00Mbit) (20.00Mbit) (20.00Mbit) (20.00Mbit)  
*** Starting CLI:  
mininet> _
```

- b) Inspeção informações das interfaces, endereços MAC, IP e portas através de linhas de comando;

```

mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s2-eth1
h3 h3-eth0:s3-eth1
h4 h4-eth0:s4-eth1
h5 h5-eth0:s5-eth1
h6 h6-eth0:s6-eth1
h7 h7-eth0:s7-eth1
h8 h8-eth0:s8-eth1
s1 lo: s1-eth1:h1-eth0 s1-eth2:s2-eth2
s2 lo: s2-eth1:h2-eth0 s2-eth2:s1-eth2 s2-eth3:s3-eth2
s3 lo: s3-eth1:h3-eth0 s3-eth2:s2-eth3 s3-eth3:s4-eth2
s4 lo: s4-eth1:h4-eth0 s4-eth2:s3-eth3 s4-eth3:s5-eth2
s5 lo: s5-eth1:h5-eth0 s5-eth2:s4-eth3 s5-eth3:s6-eth2
s6 lo: s6-eth1:h6-eth0 s6-eth2:s5-eth3 s6-eth3:s7-eth2
s7 lo: s7-eth1:h7-eth0 s7-eth2:s6-eth3 s7-eth3:s8-eth2
s8 lo: s8-eth1:h8-eth0 s8-eth2:s7-eth3
c0
mininet>

```

```

mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=1897>
<Host h2: h2-eth0:10.0.0.2 pid=1899>
<Host h3: h3-eth0:10.0.0.3 pid=1901>
<Host h4: h4-eth0:10.0.0.4 pid=1903>
<Host h5: h5-eth0:10.0.0.5 pid=1905>
<Host h6: h6-eth0:10.0.0.6 pid=1907>
<Host h7: h7-eth0:10.0.0.7 pid=1909>
<Host h8: h8-eth0:10.0.0.8 pid=1911>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=1916>
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=1919>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=1922>
<OVSSwitch s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None pid=1925>
<OVSSwitch s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None pid=1928>
<OVSSwitch s6: lo:127.0.0.1,s6-eth1:None,s6-eth2:None,s6-eth3:None pid=1931>
<OVSSwitch s7: lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None pid=1934>
<OVSSwitch s8: lo:127.0.0.1,s8-eth1:None,s8-eth2:None pid=1937>
<Controller c0: 127.0.0.1:6653 pid=1890>
mininet>

```

```

mininet> h1 ifconfig
h1-eth0  Link encap:Ethernet  HWaddr 00:00:00:00:00:01
         inet addr:10.0.0.1 Bcast:10.255.255.255 Mask:255.0.0.0
         UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
         RX packets:0 errors:0 dropped:0 overruns:0 frame:0
         TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
         collisions:0 txqueuelen:1000
         RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

lo       Link encap:Local Loopback
         inet addr:127.0.0.1 Mask:255.0.0.0
         UP LOOPBACK RUNNING MTU:65536 Metric:1
         RX packets:0 errors:0 dropped:0 overruns:0 frame:0
         TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
         collisions:0 txqueuelen:0
         RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

mininet>

```

```

s7-eth2  Link encap:Ethernet  HWaddr ea:65:ea:1b:84:a9
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s7-eth3  Link encap:Ethernet  HWaddr ba:8c:19:25:de:f2
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s8-eth1  Link encap:Ethernet  HWaddr ee:44:1d:8b:66:9e
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s8-eth2  Link encap:Ethernet  HWaddr 2a:ff:1a:89:30:87
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

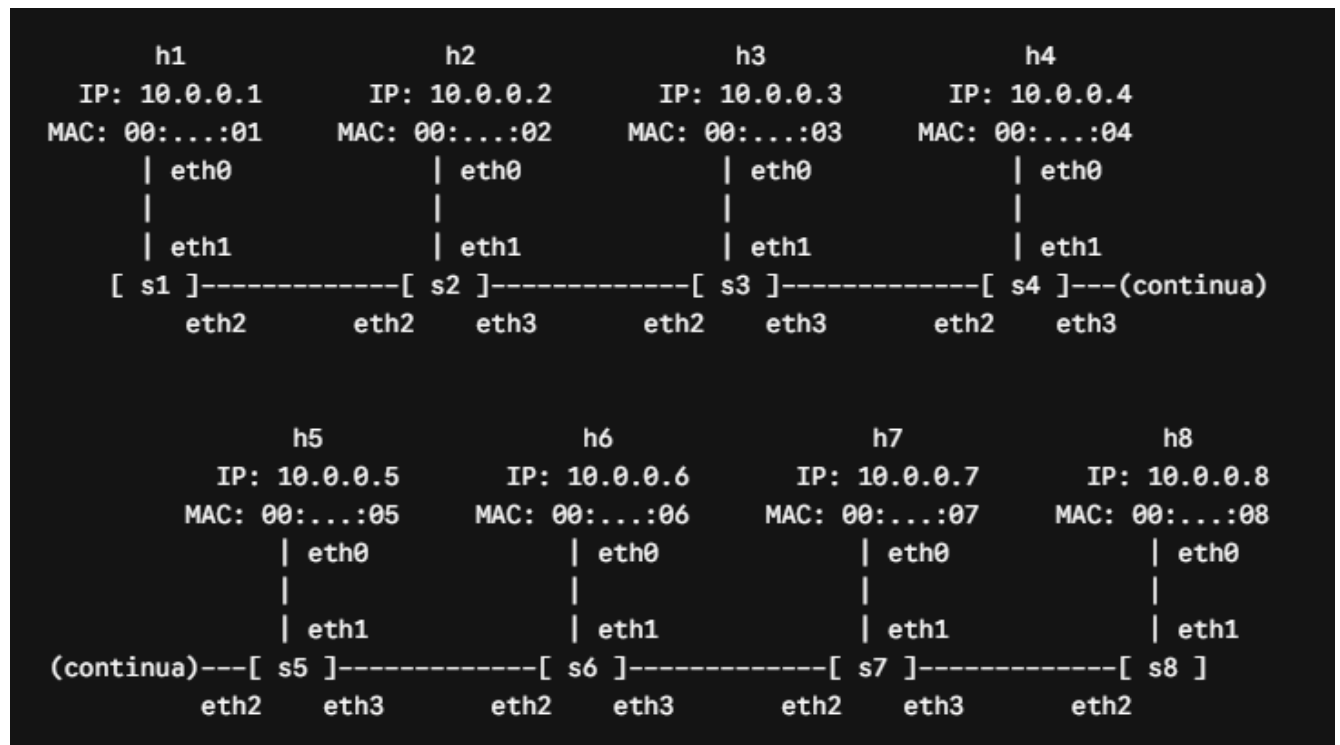
```

```

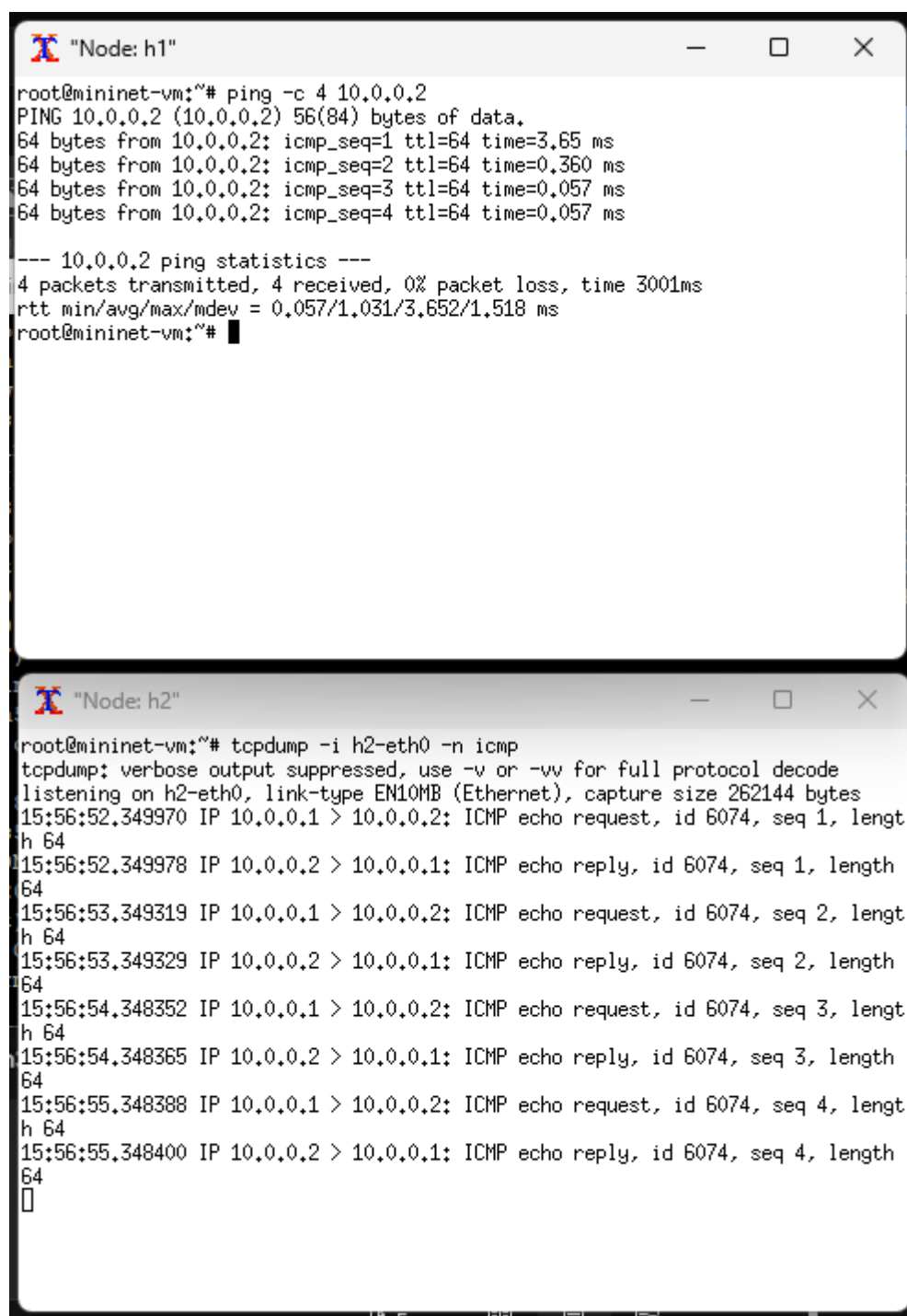
mininet> sh ovs-ofctl show s1
OFPT_FEATURES_REPLY (xid=0x2): dpid:0000000000000001
n_tables:254, n_buffers:256
capabilities: FLOW_STATS TABLE_STATS PORT_STATS QUEUE_STATS ARP_MATCH_IP
actions: OUTPUT SET_VLAN_VID SET_VLAN_PCP STRIP_VLAN SET_DL_SRC SET_DL_DST SET_N
W_SRC SET_NW_DST SET_NW_TOS SET_TP_SRC SET_TP_DST ENQUEUE
1(s1-eth1): addr:3e:88:9d:fa:41:ae
    config:      0
    state:       0
    current:     10GB-FD COPPER
    speed: 10000 Mbps now, 0 Mbps max
2(s1-eth2): addr:0a:0d:5d:87:ee:8b
    config:      0
    state:       0
    current:     10GB-FD COPPER
    speed: 10000 Mbps now, 0 Mbps max
LOCAL(s1): addr:fa:ec:66:49:8e:46
    config:      0
    state:       0
    speed: 0 Mbps now, 0 Mbps max
OFPT_GET_CONFIG_REPLY (xid=0x4): frags=normal miss_send_len=0

```

c) Crie um desenho ilustrativo da topologia com todas as informações obtidas no item anterior;



- d) Execute testes de ping entre os diferentes nós, mostre os pacotes chegando nos nós com uso do comando tcpdump.



The image shows two terminal windows. The top window, titled "Node: h1", shows the output of a ping command from root@mininet-vm to 10.0.0.2. It displays four successful ping attempts with decreasing times (3.65 ms, 0.360 ms, 0.057 ms, 0.057 ms) and a summary showing 0% packet loss and an average round-trip time of 1.031 ms. The bottom window, titled "Node: h2", shows the output of tcpdump on interface h2-eth0, capturing ICMP echo requests and replies between 10.0.0.1 and 10.0.0.2. It shows four request-reply pairs with timestamps and sequence numbers (1, 2, 3, 4).

```
root@mininet-vm:~# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.65 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.360 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.057 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3001ms
rtt min/avg/max/mdev = 0.057/1.031/3.652/1.518 ms
root@mininet-vm:~#
```

```
root@mininet-vm:~# tcpdump -i h2-eth0 -n icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on h2-eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
15:56:52.349970 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 6074, seq 1, length 64
15:56:52.349978 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 6074, seq 1, length 64
15:56:53.349319 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 6074, seq 2, length 64
15:56:53.349329 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 6074, seq 2, length 64
15:56:54.348352 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 6074, seq 3, length 64
15:56:54.348365 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 6074, seq 3, length 64
15:56:55.348388 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 6074, seq 4, length 64
15:56:55.348400 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 6074, seq 4, length 64
^C
```

- e) Especifique que o host 1 na porta 5555 vai ser um servidor TCP e o host 3 um cliente e execute testes de iperf, considere um relatório por segundo com teste de 15 segundos. Faça os testes para larguras de banda bw de 20, 30 e 40 Mbps (Necessário reconstruir a topologia para os outros valores).

Para 20 Mbps:

```

mininet> h1 iperf -s -p 5555 -i 1 &
[1] 6107
mininet> h3 iperf -c 10.0.0.1 -p 5555 -i 1 -t 15
-----
Client connecting to 10.0.0.1, TCP port 5555
TCP window size: 85.3 KByte (default)
-----
[  3] local 10.0.0.3 port 36502 connected with 10.0.0.1 port 5555
[ ID] Interval      Transfer    Bandwidth
[  3] 0.0- 1.0 sec   2.50 MBytes 21.0 Mbits/sec
[  3] 1.0- 2.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 2.0- 3.0 sec   2.38 MBytes 19.9 Mbits/sec
[  3] 3.0- 4.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 4.0- 5.0 sec   2.12 MBytes 17.8 Mbits/sec
[  3] 5.0- 6.0 sec   2.38 MBytes 19.9 Mbits/sec
[  3] 6.0- 7.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 7.0- 8.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 8.0- 9.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 9.0-10.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 10.0-11.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 11.0-12.0 sec   2.38 MBytes 19.9 Mbits/sec
[  3] 12.0-13.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 13.0-14.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 14.0-15.0 sec   2.25 MBytes 18.9 Mbits/sec
[  3] 0.0-15.1 sec   34.4 MBytes 19.0 Mbits/sec
mininet> █

```

Para 30 Mbps:

```

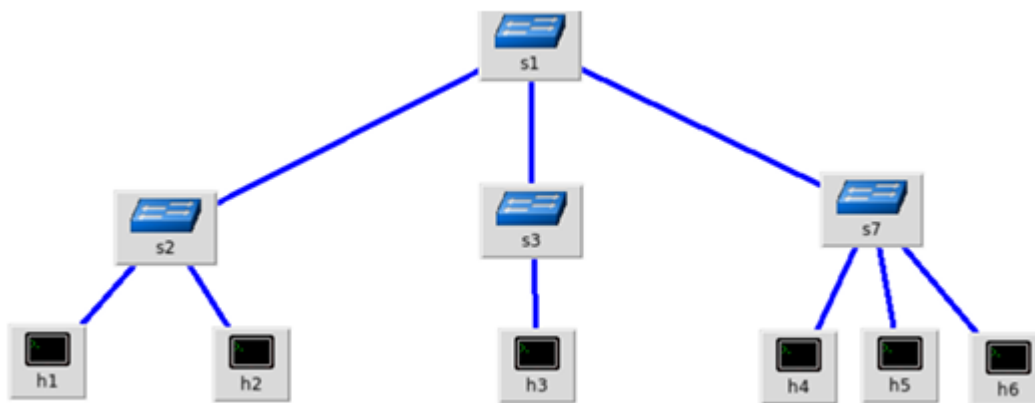
mininet> h1 iperf -s -p 5555 -i 1 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -i 1 -t 15
-----
Client connecting to 10.0.0.1, TCP port 5555
TCP window size: 85.3 KByte (default)
-----
[  3] local 10.0.0.3 port 36522 connected with 10.0.0.1 port 5555
[ ID] Interval      Transfer    Bandwidth
[  3] 0.0- 1.0 sec   3.75 MBytes 31.5 Mbits/sec
[  3] 1.0- 2.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 2.0- 3.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 3.0- 4.0 sec   3.25 MBytes 27.3 Mbits/sec
[  3] 4.0- 5.0 sec   3.25 MBytes 27.3 Mbits/sec
[  3] 5.0- 6.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 6.0- 7.0 sec   3.25 MBytes 27.3 Mbits/sec
[  3] 7.0- 8.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 8.0- 9.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 9.0-10.0 sec   3.25 MBytes 27.3 Mbits/sec
[  3] 10.0-11.0 sec   3.25 MBytes 27.3 Mbits/sec
[  3] 11.0-12.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 12.0-13.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 13.0-14.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 14.0-15.0 sec   3.38 MBytes 28.3 Mbits/sec
[  3] 0.0-15.1 sec   50.5 MBytes 28.1 Mbits/sec
mininet> █

```

Para 40 Mbps:

```
mininet> h1 iperf -s -p 5555 -i 1 &
mininet> h3 iperf -c 10.0.0.1 -p 5555 -i 1 -t 15
-----
Client connecting to 10.0.0.1, TCP port 5555
TCP window size: 85.3 KByte (default)
-----
[  3] local 10.0.0.3 port 36542 connected with 10.0.0.1 port 5555
[ ID] Interval      Transfer    Bandwidth
[  3] 0.0- 1.0 sec   5.00 MBytes 41.9 Mbits/sec
[  3] 1.0- 2.0 sec   4.38 MBytes 36.7 Mbits/sec
[  3] 2.0- 3.0 sec   4.50 MBytes 37.7 Mbits/sec
[  3] 3.0- 4.0 sec   4.50 MBytes 37.7 Mbits/sec
[  3] 4.0- 5.0 sec   4.25 MBytes 35.7 Mbits/sec
[  3] 5.0- 6.0 sec   4.50 MBytes 37.7 Mbits/sec
[  3] 6.0- 7.0 sec   4.38 MBytes 36.7 Mbits/sec
[  3] 7.0- 8.0 sec   4.25 MBytes 35.7 Mbits/sec
[  3] 8.0- 9.0 sec   4.50 MBytes 37.7 Mbits/sec
[  3] 9.0-10.0 sec   4.50 MBytes 37.7 Mbits/sec
[  3] 10.0-11.0 sec   4.00 MBytes 33.6 Mbits/sec
[  3] 11.0-12.0 sec   4.38 MBytes 36.7 Mbits/sec
[  3] 12.0-13.0 sec   4.50 MBytes 37.7 Mbits/sec
[  3] 13.0-14.0 sec   4.25 MBytes 35.7 Mbits/sec
[  3] 14.0-15.0 sec   4.50 MBytes 37.7 Mbits/sec
[  3] 0.0-15.0 sec  66.5 MBytes 37.1 Mbits/sec
mininet>
```

2. Crie um código Python para a topologia customizada abaixo:



a) Com uso de linha de comando padrão do Mininet, crie a topologia customizada considerando o endereço MAC padronizado e controlador manual;

```
mininet@mininet-vm:~$ nano custom_topo.py
mininet@mininet-vm:~$ sudo mn --custom custom_topo.py --topo mytopo --mac --controller=none
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6
*** Adding switches:
s1 s2 s3 s7
*** Adding links:
(s1, s2) (s1, s3) (s1, s7) (s2, h1) (s2, h2) (s3, h3) (s7, h4) (s7, h5) (s7, h6)
*** Configuring hosts
h1 h2 h3 h4 h5 h6
*** Starting controller

*** Starting 4 switches
s1 s2 s3 s7 ...
*** Starting CLI:
mininet>
```

b) Inspeção informações das interfaces, endereços MAC, IP e portas através de linhas de comando;

```
mininet> net
h1 h1-eth0:s2-eth2
h2 h2-eth0:s2-eth3
h3 h3-eth0:s3-eth2
h4 h4-eth0:s7-eth2
h5 h5-eth0:s7-eth3
h6 h6-eth0:s7-eth4
s1 lo: s1-eth1:s2-eth1 s1-eth2:s3-eth1 s1-eth3:s7-eth1
s2 lo: s2-eth1:s1-eth1 s2-eth2:h1-eth0 s2-eth3:h2-eth0
s3 lo: s3-eth1:s1-eth2 s3-eth2:h3-eth0
s7 lo: s7-eth1:s1-eth3 s7-eth2:h4-eth0 s7-eth3:h5-eth0 s7-eth4:h6-eth0
mininet>
```

```
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=2048>
<Host h2: h2-eth0:10.0.0.2 pid=2050>
<Host h3: h3-eth0:10.0.0.3 pid=2052>
<Host h4: h4-eth0:10.0.0.4 pid=2054>
<Host h5: h5-eth0:10.0.0.5 pid=2056>
<Host h6: h6-eth0:10.0.0.6 pid=2058>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=2063>
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=2066>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=2069>
<OVSSwitch s7: lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None,s7-eth4:None pid=2072>
mininet>
```



```

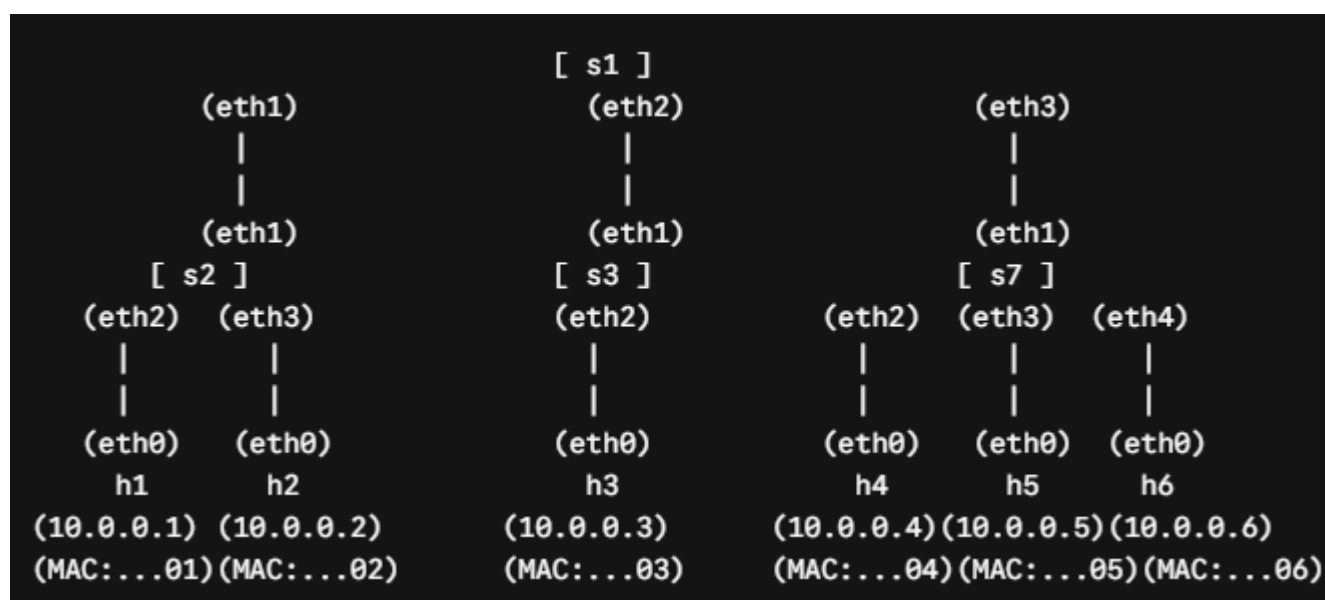
mininet> h4 ifconfig
h4-eth0   Link encap:Ethernet  HWaddr 00:00:00:00:00:04
          inet addr:10.0.0.4  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

mininet>

```

c) Crie um desenho ilustrativo da topologia com todas as informações obtidas no item anterior;



d) Faça testes de ping considerando os switches normais;

```

mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6
h2 -> h1 h3 h4 h5 h6
h3 -> h1 h2 h4 h5 h6
h4 -> h1 h2 h3 h5 h6
h5 -> h1 h2 h3 h4 h6
h6 -> h1 h2 h3 h4 h5
*** Results: 0% dropped (30/30 received)
mininet>

```

e) Apague as regras anteriores e crie regras baseadas em endereços MAC para alguns nós. (Deve-se comunicar hosts dos diferentes switches);

```
mininet> sh ovs-ofctl del-flows s1
mininet> sh ovs-ofctl del-flows s2
mininet> sh ovs-ofctl del-flows s3
mininet> sh ovs-ofctl del-flows s7
mininet>
```

Regras switch 2:

```
mininet> sh ovs-ofctl add-flow s2 dl_type=0x0806,actions=flood
mininet> sh ovs-ofctl add-flow s2 dl_dst=00:00:00:00:00:04,actions=output:1
mininet> sh ovs-ofctl add-flow s2 dl_dst=00:00:00:00:00:01,actions=output:2
mininet>
```

Regras switch 1:

```
mininet> sh ovs-ofctl add-flow s1 dl_type=0x0806,actions=flood
mininet> sh ovs-ofctl add-flow s1 dl_dst=00:00:00:00:00:04,actions=output:3
mininet> sh ovs-ofctl add-flow s1 dl_dst=00:00:00:00:00:01,actions=output:1
mininet>
```

Regras switch 7:

```
mininet> sh ovs-ofctl add-flow s7 dl_type=0x0806,actions=flood
mininet> sh ovs-ofctl add-flow s7 dl_dst=00:00:00:00:00:04,actions=output:2
mininet> sh ovs-ofctl add-flow s7 dl_dst=00:00:00:00:00:01,actions=output:1
mininet>
```

f) Faça testes de ping para demonstrar que as regras foram bem implementadas.

H1 pingando H4: sucesso pois temos a rota do MAC ...:01 até o ...:04

```
mininet> h1 ping -c 4 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=0.977 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.088 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.086 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=0.097 ms

--- 10.0.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 0.086/0.312/0.977/0.383 ms
mininet>
```

H1 pingando H2: pacote derrubado, provando as regras baseadas em MAC estão funcionando

```
mininet> h1 ping -c 4 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3024ms

mininet>
```